

MEHEK MANDAL

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AI agent engineer and founder. Sole engineer of the G42 Intelligence Agent Platform submitted with this application: designed, built, benchmarked, and deployed end to end.

SELECTED SYSTEMS

G42 Intelligence Agent Platform · this submission

2026

- Architected a governed agent chassis (LangGraph, FastAPI) serving 3 enterprise agents (Finance, Human Capital, Strategic Sourcing) from 1 live deployment: hash-addressed config snapshots, replayable checkpoint audit trail, OpenTelemetry, Presidio PII redaction, graph-level JWT RBAC, per-task budget rails.
- Benchmarked with a purpose-built oracle-anchored eval harness: 135/135 run reliability, escalation precision 1.00 on all 3 agents, 10/10 identity-pair bias parity (Human Capital), 10/10 adversarial attack classes defended, \$0.0006 average cost per task.
- Engineered a governed self-improvement loop (eval, diagnose, propose, shadow eval, human gate) that lifted finance task accuracy from 67% to 100% across 2 promoted revisions, each recorded in the CHANGELOG with its benchmark delta.

Nucleus at Elemental Growth, AI agency · Founder

2025 - Present

- Built and operate Nucleus, the agency's production analytics and client-delivery pipeline supporting growth engagements with Perplexity, Gamma, and Chess.com.
- Designed its 15+ integration connector architecture (fetch/normalize/cache); reused the same pattern as the connector layer of the G42 platform.

Dial · LLM routing and evaluation

2026

- Developed an eval harness with a 9-metric scorecard: oracle labeling via batch inference, LLM-as-judge with anti-bias controls (position swap, self-consistency), cache-aware cost modeling; adapted as the G42 platform's benchmark methodology.

EXPERIENCE

Traeco, AI agent cost infrastructure · Co-Founder and CEO

Feb 2026 - Jun 2026

- Ran 50+ enterprise discovery interviews to map agent cost and governance failure modes; translated findings into the SDK architecture and product roadmap.
- Engineered the Python SDK, LangChain instrumentation layer, and LLM-as-judge eval pipeline (GPT-4o, Claude Sonnet) quantifying cost-quality deltas on every agent change before deployment.
- Deployed to 3 enterprise design partners in 6 weeks; built live traffic replay and config-layer observability (configuration captured at onboarding, not trace-only), now the audit-snapshot pattern in the G42 platform.

Tonee, AI texting tone detector · Founder

Sep 2025 - Present

- Shipped real-time tone detection to 100+ active users in 8 weeks, owning the full stack from model fine-tuning through iOS deployment.
- Improved workplace tone accuracy from 78% to 89% by collecting 8,300+ message annotations across 47 user interviews and iterating training data 3x with active learning.
- Cut inference latency from 2.3s to 0.1s by profiling the API call stack, isolating the round-trip root cause, and executing a Core ML on-device migration.

SoFi · Product Management Intern

Feb 2025 - May 2025

- Analyzed the onboarding funnel for 80K monthly sign-ups; identified the verification step as the 40% drop-off point from 50+ interview transcripts and behavioral data.
- Delivered a research brief documenting 2 critical friction points; the PM team's implemented changes contributed to a 15% retention improvement.
- Built a tracking plan for 4 Amplitude behavioral events on the verification flow and validated data integrity with engineering, enabling the root-cause funnel analysis.

LEADERSHIP

Spark SC · VP of Finance and Sponsorships

May 2025 - Present

- Secured \$14K across 7 sponsors and managed a \$27K budget for 12 events; improved sponsor renewal from 40% to 71% year over year.

Venture Capital Academy · President

May 2025 - Dec 2025

- Led USC's first student VC program for 130 students; co-designed an 8-week curriculum and coordinated Bay Area visits to Lightspeed, NEA, and Altos Ventures for 25 students.

EDUCATION

University of Southern California, Viterbi School of Engineering, Los Angeles, CA

May 2027

BS Computer Science and Business Administration. GPA 3.89/4.0. USC Lava Lab startup accelerator (16-week cohort).

SKILLS

Python, TypeScript, LangGraph, LangChain, FastAPI, Docker, OpenAI API, Anthropic SDK, agent orchestration, LLM evaluation, fine-tuning (LoRA/QLoRA), RAG, OpenTelemetry, SQL